

Relations

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Overview

Sequences (4.2)

Binary Relations (4.4)

Definition

Definition: A *sequence* is an ordered list of items.

Like a set: Collection of items.

Unlike a set: Order matters, can have repeats.

Simple example: Cartesian coordinates like (x, y) .

Definition: The *empty sequence* is a sequence of length 0 and is denoted λ .

Operation on sets: Product

Definition: The product of two sets A and B is written $A \times B$ and is the set of all length-two sequences where the first element of the sequence comes from A and second from B .

Example:

- ▶ $X = \{1, 2\}$
- ▶ $Y = \{0, 2, 4\}$
- ▶ $X \times Y = \{(1, 0), (1, 2), (1, 4), (2, 0), (2, 2), (2, 4)\}$
- ▶ $|X \times Y| = 6 = |X| \times |Y|$

Extends to chains of products: $S_1 \times S_2 \times S_3 \times \cdots S_k$.

If $S_1 = S_2 = S_3 = \cdots = S_k = S$, can write simply S^k .

Example: $\{a, b\}^3 = \{(a, a, a), (a, a, b), (a, b, a), (a, b, b), (b, a, a), (b, a, b), (b, b, a), (b, b, b)\}$.

Binary relations, intuitively

Binary here just means “two”. Relation means we’re talking about a relationship between two items.

Examples:

- ▶ “less than”, $a < b$ defines a relationship between two items. The relationship *holds* if and only if a represents a smaller number than b . $5 < 11$, $2021 \not< 1984$.
- ▶ “subset”, $A \subseteq B$.
- ▶ “is the square of”.
- ▶ “is a factor of”.

Binary relations, formally

Definition. A *binary relation*, R , consists of a set, A , called the domain of R , a set, B , called the codomain of R , and a subset of $A \times B$ called the graph of R .

Relation notation

$R : A \rightarrow B$. We say R maps A to B .

Kind of like a two-place predicate.

If the relation is from a set A to itself, we say it's a relation "on A ".

Infix notation is common: $a R b$, meaning (a, b) is in the graph of R .

Binary relation examples

Binary relations are essentially functions that take in two items and return a Boolean.

- ▶ superset ($\supset$)?. Yes.
- ▶ or? Sure, you could see it that way.
- ▶ add (+)? No, binary, but returns a number.
- ▶ equals (=)? Yes.
- ▶ is divisible by? Yes.
- ▶ element of ($\in$)? Yes. Domain and codomain can be different.
- ▶ is a perfect square? No, only takes one input. It's a predicate.
- ▶ is the perfect square of? Yes.
- ▶ greatest common divisor? No, returns a number.
- ▶ greatest common divisor is one? Yes, also called “relatively prime” or “co prime”.

Other binary relations and problems

- ▶ Is married to?
- ▶ Is taller than?
- ▶ Is older than?
- ▶ Happened before?
- ▶ Has met?
- ▶ Is faster than?
- ▶ Is smarter than?
- ▶ Lives with?
- ▶ Speakers the same languages as?
- ▶ Speaks a same language as?